LEARNER GUIDE

Stretch and reflect

Quadratics. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Last lesson you built y = x^2 and met its U-shape, the parabola. Today you ask one question: what happens when you put a number in front of the x^2? That number is called a coefficient, and we write the rule as y = ax^2. The quickest way to feel it is a table: for every x, the height of y = 2x^2 is simply twice the height of y = x^2, and the height of y = ½x^2 is half of it. So a stretches the curve. When a is bigger than 1 the curve is pulled tall and narrow; when a is between 0 and 1 it is pushed wide and flat; and when a is negative the whole curve is reflected — it flips upside-down into a ∩. Through all of this the turning point stays at (0, 0), because a × 0 is always 0. You will build the family with a table, move a live dial to stretch and flip the curve, read a curve's steepness to name its a, and — as a stretch — find the a that makes the curve pass through a chosen point. There is no calculator today. The question you are exploring is: what does the number in front of x^2 do to the curve?

What you are learning to notice

- That in y = ax^2, every height is the y = x^2 height multiplied by a.
- That a bigger a makes a narrower, steeper curve, and an a between 0 and 1 makes a wider, flatter one.
- That a negative a reflects the curve, so it opens downward into a ∩.
- That changing a never moves the turning point (0, 0) or the line of symmetry.

Moving through it, one step at a time

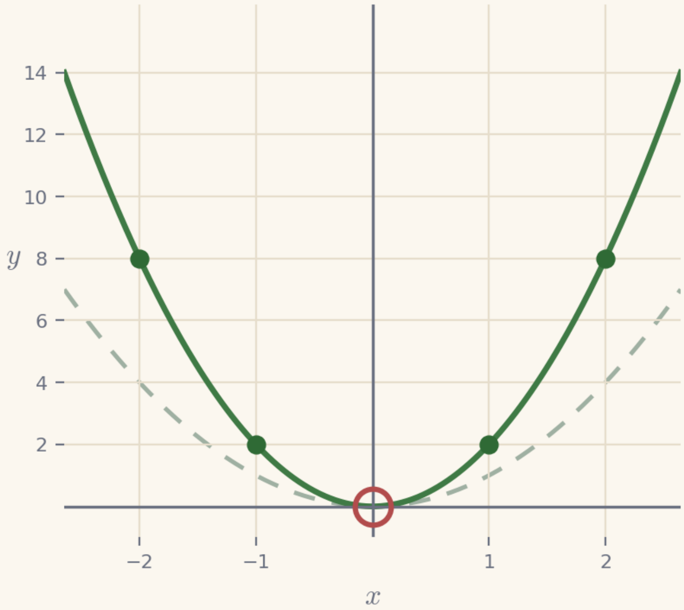
In Same curve, scaled, fill a table for y = 2x^2 and y = ½x^2 and plot all three curves together. In The a-dial, move the dial and watch y = ax^2 stretch, flatten and flip against a still copy of y = x^2. In Read the steepness, name the a that produced a drawn curve. In Make it pass through a point (a stretch), work out a = y ÷ x^2 so the curve goes through a chosen point.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — stretch it — y = $2x^2$

Plot y = $2x^2$ by scaling y = x^2. For each x, work out x^2 and then multiply by 2. The dashed curve is y = x^2 for comparison.

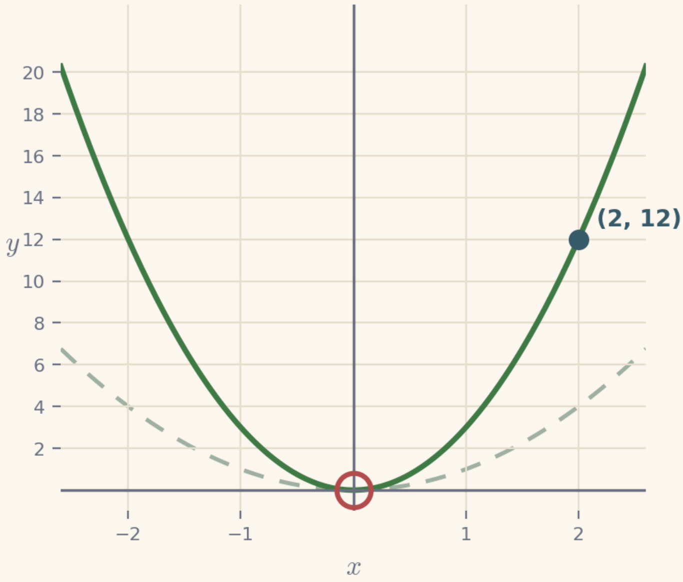


1. $x = 1$	$2 \times 1^2 = 2$
2. $x = 2$	$2 \times 2^2 = 8$
3. compare with x^2	every height is doubled

y = $2x^2$ is narrower and steeper than y = x^2, but shares the turning point (0, 0)

Example 2 — make it pass through a point

Find a so that $y = ax^2$ passes through (2, 12). Put the point into the rule and solve for a.



1. the rule — $y = ax^2$
2. put in the point — $12 = a \times 2^2$
3. so — $12 = a \times 4$
4. divide — $a = \dfrac{12}{4} = 3$

a = 3, so $y = 3x^2$ passes through (2, 12)

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (plotting $y = x^2$ (Lesson 5), multiplying by a whole number and by a half, and squaring including squaring a negative), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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